



Toni Blair

Software Engineer | Full-Stack & Backend Developer
4th year Dual Degree Student | IIT Kharagpur



Email



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EDUCATION

2023 - present	Indian Institute of Technology, Kharagpur	7.4	Dual Degree: Ocean Engg & Naval Arch
2023	Tata DAV school, Jamadoba(DHN)	83.30%	12th CBSE Board
2021	Denobili School F.R.I, Dighwadih	85.00%	10th ICSE Board

SKILLS AND EXPERTISE

Languages: Python, JavaScript, C++, SQL | **Frontend:** React.js, React Native, HTML5, CSS3, Tailwind CSS, Bootstrap | **Backend:** FastAPI, Node.js, Express.js, REST APIs, WebSockets, JWT Authentication | **Databases:** PostgreSQL, SQLite, Redis | **Tools:** Git, GitHub, Docker, Linux, VS Code, Postman | **Concepts:** Data Structures & Algorithms, OOP, DBMS, REST API Design

PROJECTS

- Chalo ---- Hyperlocal Activity Platform** | *React Native, FastAPI, PostgreSQL, Async SQLAlchemy, JWT, Docker*
- Building a mobile platform for discovering and joining nearby activities, with a React Native frontend and FastAPI backend organized around separate application modules.
 - Implemented backend foundations including JWT-based authentication, asynchronous database workflows with PostgreSQL/SQLAlchemy, OTP generation and expiry handling, and API endpoints for user/activity workflows.
 - Designed the project with a modular client-server architecture and Dockerized backend setup; actively extending features including verification, onboarding and activity discovery.
- Wyrn Store --- Full-Stack E-Commerce Platform** | *React (Vite), Node.js, Express.js, MongoDB, JWT, Cloudinary*
- Built an end-to-end e-commerce application with secure authentication, role-based admin controls, product catalog, cart, checkout and order-management workflows.
 - Developed REST APIs for authentication, products, carts, orders and administrative operations, including user/role management, order status updates and sales analytics.
 - Implemented image upload support through Cloudinary and built protected frontend routes for customer and
- Multithreaded HTTP Server** | *C, POSIX Sockets, pthreads, HTTP, Linux, Make*
- Built a lightweight HTTP server from scratch using TCP sockets and POSIX threads, accepting client connections and handling requests concurrently with a thread-per-client model.
 - Implemented manual HTTP request parsing and response construction, static file serving, Content-Type handling and 404 responses.
 - Created a Makefile-based build workflow and documented architecture and current limitations, including the next-step improvement of introducing a bounded thread pool.

POSITIONS OF RESPONSIBILITY

- Design Sub-Head | Samudramanthan (OENA Dept Fest), IIT Kharagpur** November '24 - present
- Led the design vertical for the OENA departmental fest; created posters, banners, ID cards, certificates and digital creatives.
- Maintainance & Socult Secretary | Rajendra Prasad Hall, IIT Kharagpur** August '24 – May '25
- Coordinated hall maintenance and student welfare; contributed to hall merchandise and design initiatives.

LEADERSHIP & ACHIEVEMENT

- Gold Medal (Team) --- Sketching Event, Inter-Hall General Championship, IIT Kharagpur.
- Design Team Member --- Finess (Fashion Design), Inter IIT Cultural Meet 7.0.
- NSO Coordinator --- Hockey & Health and Fitness, IIT Kharagpur.